A Project Report on

E-commerce Web Application using Machine Learning

By

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List of Abbreviations and Acronyms

AI = Artificial Intelligence.

B2C = Business-to-Customer or Consumer

NLP = Natural Language Processing.

MCDM = Multivariate comparative machine learning method.

CUSM = Covenant University Shopping Mall

PCA = Principal component analysis.

BERT = Bidirectional encoder representations from transformers.

ML= Machine Learning.

UX = User Experience.

SDLC = Software Development Life Cycle.

Declaration

I hereby declare that the project entitled E-Commerce Web Application using Machine Learning was submitted for the degree of Bachelor of Science in Computer Science and Engineering at the Bangladesh University of Business and Technology (BUBT). This project is our original work and does not contain any previously accepted material for other degrees or diplomas, except as noted in the project's next section. It also does not include any previously published or written materials.

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Approval

I do hereby declare that the research works presented in this project, entitled, "E-Commerce web application using Machine Learning", is the outcome of the original works carried out by Sunjida Sultana Nisha, Latiful Islam Shoisob, Nahid Hossain Shohag, Meherin Afrin, Panna Akter under my supervision. I further declare that no part of this project has been submitted elsewhere for the requirements of any degree, award diploma, or any other purposes except for publications. I certify that the dissertation meets the requirements and standards for the Bachelor of Science in Computer Science and Engineering degree.

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Certification

This is to certify that the report on the ”E-Commerce Web Application using Machine Learning” has been carried out by Sunjida Sultana Nisha(ID:1920213005), Latiful Islam Shoisob(ID:19202103162), Md. Nahid Hossain Shohag(ID:19202103181), Meherin Afrin(ID:19202103192), and Panna Akter(ID:19202103194) of Bangladesh University of Business and Technology (BUBT) as a partial fulfillment of the requirement of practicum defense course. The report, written under my direction, is a record of the completed task carried out effectively. As far as I know and following their statement, no portion of this report has been sent in for any kind of certification, degree, or recognition. Now they are ready to submit the report. I hope whatever they do in the future turns out well.

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Abstract

"GOHONA" is an e-commerce web app for jewelry buyers, shoppers and retailers, offering easy browsing, purchase, and data management. It features two modules: customer and admin. Our system offers a user-friendly shopping experience.

The system is developed by using Python, HTML, CSS, JS, and Django framework. And also we deploy a recommendation model to our website for product suggestions. Our system helps customers who want to buy exclusive jewelry in a user-friendly manner. Customers can simply visit the system, choose products from any category, register for an order, log into their accounts, and finally make payment of their purchased amount. The customers can edit their order list before order confirmation. It provides 24x7 support. Customers can make inquiries about any products or services and place orders anytime, anywhere from any location. He/she has to pay by SSLCOMMERZ or COD and will get home delivery.

Overall our system "GOHONA" is an exciting website. We are happy to build this project and we have enjoyed it.

Chapter 1

1 Introduction

1.1 Introduction

Introducing our cutting-edge e-commerce platform, which combines advanced technology with a user-friendly design. Our platform is equipped with machine learning to ensure your shopping experience is protected. With a strong emphasis on fraud detection, each transaction is thoroughly analyzed to guarantee the highest level of security. The user-friendly interface facilitates browsing and purchasing, allowing you to explore a wide variety of products with ease. Additionally, an intelligent system is integrated to provide immediate assistance, improving your shopping experience from start to finish. At the perfect combination of cutting-edge technology and user-friendly design, we are ready to revolutionize the way you shop online, providing a secure, interactive, and dynamic marketplace.

1.2 Project Motivation

1. Personalised Recommendations: By providing customised product recommendations based on user behaviour and preferences, it may enhance user experience and boost sales.

2. Dynamic Pricing: Evaluate demand, competitors, and consumer behaviour to optimise pricing strategies in real-time to maximise revenue and profitability.

3. Predictive Analytics: Increase the accuracy of sales forecasting and inventory management with predictive analytics to maximise stock levels and reduce stockouts.

4. Fraud Detection: By applying machine learning algorithms to identify questionable activity in transactions, it may improve security and guard against fraud detection.

5. Sentiment Analysis: Use product reviews and feedback to gain insightful information about customer sentiment that can be used to better marketing campaigns and product offerings.
6. Market Basket Analysis: Analyzing and examining consumer buying trends, find chances for product bundling and cross-selling to raise average order value.

1.3 Project background

The e-commerce industry has been moving into exponential growth in recent years, with an exponential number of businesses transitioning to online platforms to reach a global audience. However, the competitive landscape presents numerous challenges, including the need for personalized user experiences, dynamic pricing strategies, efficient inventory management, and robust fraud prevention systems This project's main goal is to create an online e-commerce platform with machine learning features integrated. The project's specific objectives are to:

1. Enhance the user experience by providing personalized product recommendations tailored to individual preferences.
2. Implement dynamic pricing techniques to increase sales and profit margins.
3. Improve inventory management and sales forecasting accuracy.
4. Enhance security through the detection of fraudulent activities in transactions.
5. Analyze customer sentiment from product reviews and feedback to inform product improvements and marketing strategies.
6. Identify opportunities for cross-selling and product bundling to increase the average order value.

1.4 Key Feature

The e-commerce landscape is characterized by a growing demand for seamless online shopping experiences it provide e-commerce solution and services to customer using some integrated tools like product display tools, shopping cart, payment systems, etc. This will have the following key features:

1. An online store displaying a wide range of electronic products for easy purchase.
2. A dedicated website that makes it simple for clients to compare the costs and quality of products.
3. Various payment methods, including pay-after-delivery, credit card, and mobile payment.
4. An easy-to-use shopping cart feature that enables customers to add, remove, and manage chosen products in their shopping carts.

1.5 Expected Impact

- B2C E-commerce: The establishment of the online store signifies the start of business-to-consumer e-commerce in our nation, providing new channels for companies to communicate with clients directly.
- Encouragement for Other Sectors: This project intends to encourage other sectors of our nation to enter the e-commerce space by offering a template or generic function development.By utilising the platform's concept, they can design their own unique e-commerce portals that are tailored to their particular requirements and sector.
- Increasing economic growth: By facilitating trade, improving market access, and generating employment possibilities, the spread of e-commerce across multiple industries has the potential to increase economic growth.
- Increased Consumer Convenience: Since internet buying platforms are readily available, consumers may access information more easily and generating job possibilities.
- Digital Transformation: Businesses that embrace e-commerce are more likely to implement digital processes and technology, which promotes innovation and market competitiveness.

1.6 Report Breakdown

The report begins with a review of existing literature on e-commerce and the readiness of developing countries to establish theoretical content.This design of the e-commerce portal is outlined, detailing its features and structure as informed by the literature review and project objectives. Subsequent sections cover the testing phase, assessing the platform's functionality and user experience through various test cases.Finally, the report concludes by summarizing key insights and recommendations drawn from the literature review, portal design, testing, and findings, while also suggesting avenues for future research or improvement.

Chapter 2

2 Survey of Existing model

2.1 Background of online e-commerce in Bangladesh

Bangladesh's online e-commerce has grown significantly in the last several years, coinciding with a rise in smartphone adoption nationwide. There are several potentials in this growing sector for developing machine learning-enhanced e-commerce web applications. These applications might be completely transformed by machine learning, which would allow for more efficient customer service procedures, improved fraud detection systems to protect transactions, and customized suggestions based on the interests of each individual user. However there are other obstacles to e-commerce in Bangladesh, including a large segment of the population lacking basic digital literacy and logistical difficulties. The effective deployment and broad uptake of machine learning solutions in e-commerce platforms will depend on overcoming these obstacles, which will guarantee continued growth and competitiveness in this fast-paced industry.

2.2 Statistical data system

A machine learning-capable web application for statistical data e-commerce requires several important phases to be built. To start with, you set the requirements, identifying the types of statistical data you want to examine, including trends in sales, consumer behavior, or product recommendations. Then, you gather pertinent information from your e-commerce platform, such as past transactions, client profiles, and product specifications. After the data is collected, it is processed, cleaned, and prepared for analysis by addressing any missing values. The next step is to perform exploratory data analysis to identify patterns and trends that will help you choose features. Feature engineering is the next step, in which unprocessed info is transformed into meaningful features that are used as input by machine learning algorithms. These algorithms—which include clustering for data mining and regression for sales forecasting—are chosen according to your particular use case.

2.3 Problem in the existing approach

The low recommendation accuracy of the current machine learning-based e-commerce web application is a major problem. Even while the recommendation system has the ability to improve user experience and increase revenue, it frequently promotes products that are unpopular or unnecessary. An insufficient amount of training data or old techniques may be the cause of this. Users could get upset or bored as a result, which would harm the platform's brand and earnings. Retraining models with new data, updating recommendation techniques, and enhancing data quality are just a few of the complete strategies needed to address this challenge. Maintaining high accuracy and relevance over time requires constant monitoring and improvement of the system.

Author Name	Methods	Focused Area	Lacking
Victoria Oguntosin[4]	Natural language process (NLP)	Chatbots can be used in a variety of circumstances, including e-commerce and non-e-commerce settings	Pattern-based chatbots have limited intelligence, weak AI capabilities, and cannot effectively handle complex queries
A.R.D.B Landim[1]	Recurrent Neural Networks (RNN), Convolutional Neural Networks (CNN), Natural Language Processing (NLP)	Chatbot design and comprehensive map for fashion e-commerce using NLP, CNN, RNN	Lack of technology immaturity and privacy issues. Models that measure consumer acceptance should be addressed also considering the difference in women's wear and men's wear categories.
Anthony Raj Amaranathan et al.[6]	Principal Component Analysis (PCA), K-means Clustering, Machine Learning for Recommendations	developing an image-based recommendation system using innovative machine learning methods	Don't provide information about the dataset used or the nature of the images (e.g., product images, user-generated images, etc.), It doesn't elaborate on how PCA and K-means clustering is applied in detail.

Table 1: This table compares the limitations and performances of existing approach

2.4 Proposed solution to problems

When machine learning solves these important issues, e-commerce can be improved. Sales are increased by personalized recommendations, revenue is maximized by price dynamics, and security is improved by fraud detection. Inventory control keeps products out of problems, and chatbots improve customer service. Semantic search improves results, targeted marketing increases engagement, and feedback analysis inspires product improvement. E-commerce systems may improve consumer satisfaction, raise revenue, and optimize operations using machine learning. Ultimately, companies may boost productivity, maximize performance, and provide outstanding customer experiences through using machine learning in each part of their online business.

Chapter 3

3 Proposed model

3.1 Introduction

Our E-commerce web application's suggested model aims to improve the platform's predictive analytics, customer segmentation, and product recommendation by utilizing machine learning approaches. Our goal is to optimize business operations and maximize income by utilizing machine learning to deliver a personalized and effective shopping experience for users.

- Recommendation System: We plan to implement a recommendation system that suggests products to item-based on which products they click to show. This system will utilize collaborative filtering techniques to identify similar items and recommend items that they have shown interest in.
- Chatbot for Customer Support: We will implement a chatbot using Natural Language Processing (NLP) techniques and conversational AI technologies. The chatbot will be accessible to users via a chat interface embedded with the e-commerce platform. The chatbot will be trained based on customer queries and support tickets to ensure accurate responses. The chatbot will handle different types of customer interactions including product inquiries, order status updates, troubleshooting assistance, and return and refund processing. If any user is having problems to order something the chatbot will help them connect with the human customer support executive. Integration with backend systems and APIs will enable the chatbot to access real-time data such as inventory status, order history, and customer profiles.
- This seamless integration will enable the chatbot to provide personalized and relevant assistance tailored to each user's context and history. The chatbot is very helpful for a website because a human cannot be active 24/7 and also

cannot do nonstop work. An AI chatbot can do non-stop work day by day.

- Continuous Improvement and Iteration: It is important to note that our proposed model is not static but rather iterative and adaptable. We will continuously monitor model performance, gather user feedback, and incorporate new data and features to improve the accuracy, effectiveness, and relevance of our machine-learning models. Regular updates and iterations will ensure that our E-commerce platform remains competitive, responsive to user needs, and aligned with evolving market dynamics.

3.2 E-commerce can be divided into primarily four categories

Technical Feasibility: The proposed e-commerce platform with integrated fraud detection, a user-friendly interface, and an interactive chatbot is evaluated for its technical feasibility in terms of the technology requirements for its effective implementation. The technological issues, problems, and solutions covered in this part are what support the project's feasibility.

User Interface Design: User experience (UX) design, frontend technologies, and web development know-how are all necessary to produce an engaging and intuitive user interface. The choice of frontend frameworks, responsive design guidelines, and seamless integration with backend components are all included in the feasibility analysis. It may be difficult to provide aesthetically appealing design elements, optimize page load times, and ensure cross-browser compatibility.

Chatbot Integration: Natural language processing (NLP) strategies and chatbot frameworks must be thoroughly understood in order to integrate a chatbot into an e-commerce platform. The feasibility analysis includes selecting an appropriate chatbot platform, teaching it to comprehend user inquiries, and creating dialogue flows that provide pertinent responses. Key technological issues include ensuring real-time interaction, user context retention, and easy connectivity with the website's backend.

Data Security and Privacy: The deployment of severe data security measures, such as the encryption of sensitive user data and safe data transmission during transactions, is technically feasible. Building user trust requires observing data protection laws and making sure privacy standards are satisfied.

3.2.1 Economic Feasibility:

The established e-commerce project's financial viability is examined, taking into account development expenses, possible profits, and long-term sustainability. The objective of this study is to give users a thorough understanding of the project's economic effects.

Development Costs: The early phase of the project entails costs for infrastructure setup, software development, and human resources. To create the machine learning model, design the user-friendly interface, and include the chatbot.

Infrastructure and Operational Costs: Infrastructure and operational investments must be made continuously to keep the platform running. In order to guarantee the best speed and security, this also includes server upkeep, database management, hosting costs, and monthly updates. The price of customer care, training, and user support should also be taken into account.

3.2.2 Operational feasibility:

Operational feasibility evaluates how feasible it is for our organization to establish and manage the suggested e-commerce website. To ascertain whether the project is compatible with your organization's resources, capabilities, and processes, entails examining many different criteria. Here are some important things to think about:

User-Friendly Interface: The website's user-friendly interface supports your business's objective of giving clients a seamless buying experience. An interface that is simple to use, visually beautiful, and intuitive can be designed and developed.

Security Measures: Our organization's dedication to protecting customer data and combating fraud is aligned with the implementation of fraud detection and security measures.

Customer Support: It is possible to integrate a chatbot and customer support system, enabling customers to get help and have their questions answered in real-time.

In summary, the proposed model for our E-commerce web application using ML represents a user-friendly experience, driving business growth, and mitigating risks. By integrating these machine learning capabilities into our platform, we aim to deliver personalized recommendations, targeted marketing strategies, and actionable insights that empower both users and stakeholders alike.

Chapter 4

4 User Manual

4.1 Introduction

In this chapter, we will discuss in detail the process of designing the e-commerce web app. We will look at the project methodologies used as well as the models of the system. Our approach to achieve the goal of a usable E-commerce system. This approach was selected as it will reduce the cost and implementation of the system.

4.1.1 The Project Methodology

We will first gather requirements and then analyze the feasibility. Then we planned for our project. After we go for the development, the development has a two-step frontend and backend. The next process will go for data collection from the API. The next process will be for Data Split. After data split we found train data and test data. In Train Data we used KNN and RMSE and after training the algorithm it will be ready for prediction using test data. After prediction we got results for related product recommendations.

4.1.2 Process Workflow Diagram

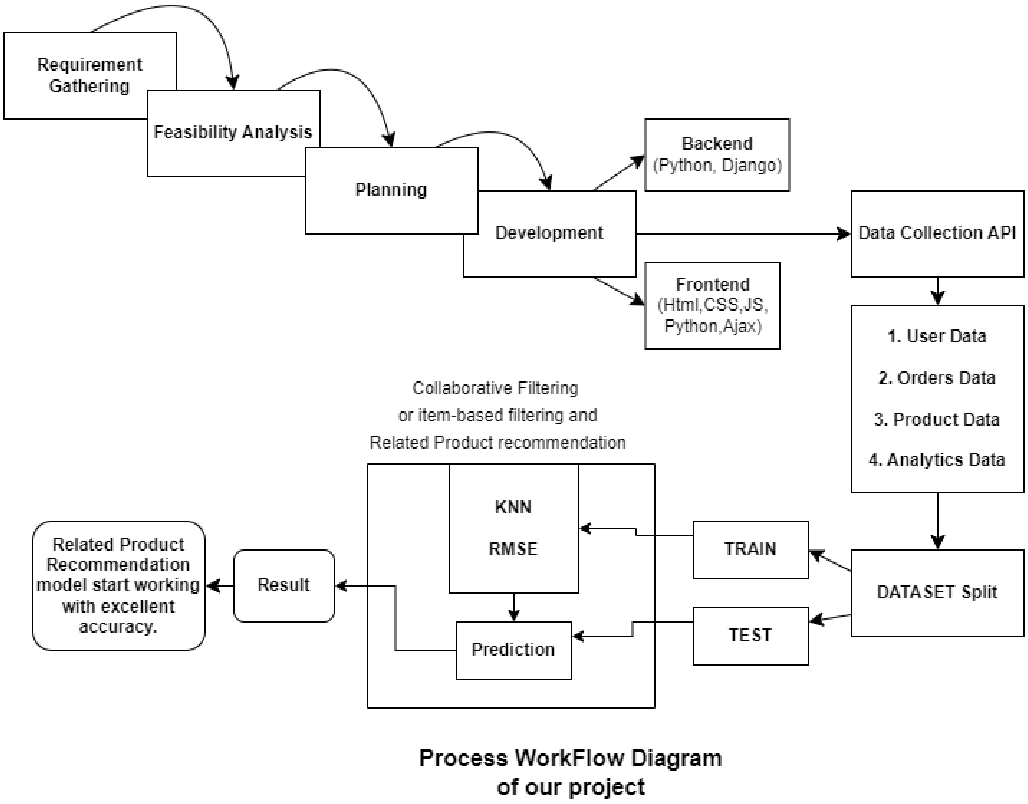


Figure 4.1: Process workflow diagram of e-commerce web application using ML

4.1.3 Project ER Diagram

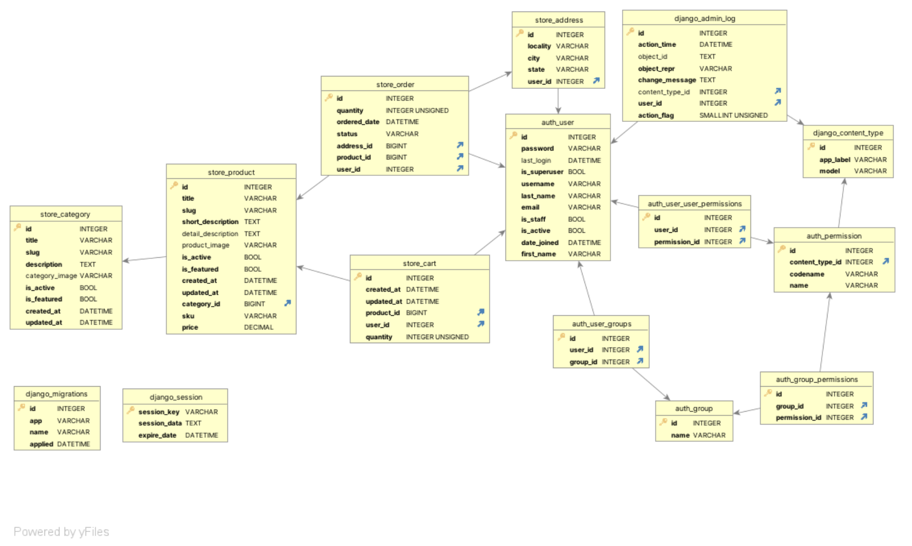


Figure 4.2: ER Diagram of our E-commerce Application

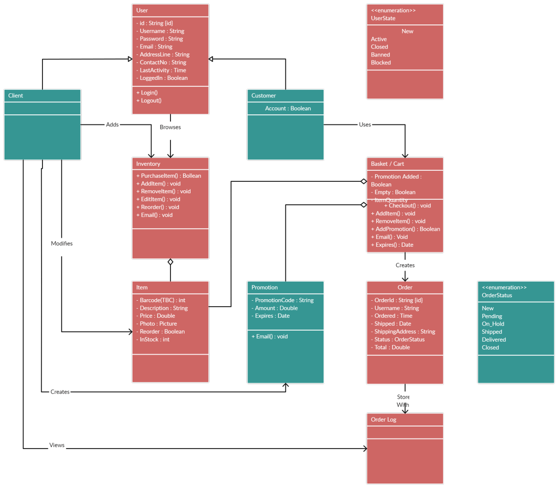


Figure 4.3: Class diagram of E-commerce web app

4.1.4 Project Flowchart

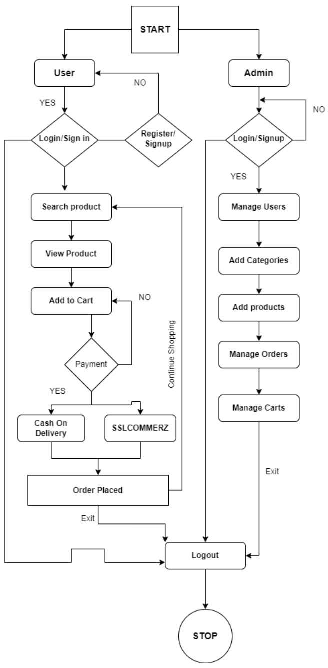


Figure 4.4: Project Flowchart Diagram

Features of our Ecommerce website

For USER:

- Home page
- Registration
- Login
- View Products
- Add to cart
- Manage Orders
- Make Payments
- Checkout
- Manage profile
- Change password

For ADMIN

- Create Superuser
- Admin Login
- Admin Dashboard
- Manage Users
- Manage actions
- Add Products
- Delete Products
- Manage Orders
- Manage cart
- Remove users
- Manage Payment

4.2 Software Development Life Cycle

The Agile SDLC Model is going to be used for the software development life cycle (SDLC) of e-commerce web applications.

Why did we choose the Agile model?

The main reason for choosing this model is that-

- Team members need to decide on a pace that will allow them to work efficiently and consistently deliver useful software. The main goal of this approach, once implemented, is to avoid professional burnout and the need for extra effort.
- Process optimization is the answer. Components, their interfaces, and their actions are defined by architecture. Architecture is the deliverable.
- The strategy for meeting the requirements is defined in the design document. This stage is known as the "how" phase.
- Engineering details are defined for computer programming languages or environments, hardware or software, packages or applications architecture, distributed architectures layers, memory size or platform, algorithms, data structures or global type definitions, interfaces, and many other topics.
- The design may make use of pre-existing components.

The stages of using the Agile method include:

- **Requirement Gathering:** In this state we collect data from the existing e-commerce website and projects and evaluate them.
- **Design the Requirements:** According to the user's requirement, We can obtain and illustrate the estate for our system. In this state, we have designed and developed the e-commerce website for corporate sector diagrams such as the Class diagram, and activity diagram, and made a prototype of our overall project.
- **Testing and quality control:** In this stage different types of testing strategies such as function testing unit testing security testing performance testing or

applied to our project. We have drawn our DFD-level diagrams with these things in mind.

- **Deployment:** During these face, we have implemented our e-commerce website for the corporate office in Bangladesh according to our plan.

- **Feedback:** This is the last phase of the process before the product is made available to the public. This team gathers feedback on the results and compiles it into a report.

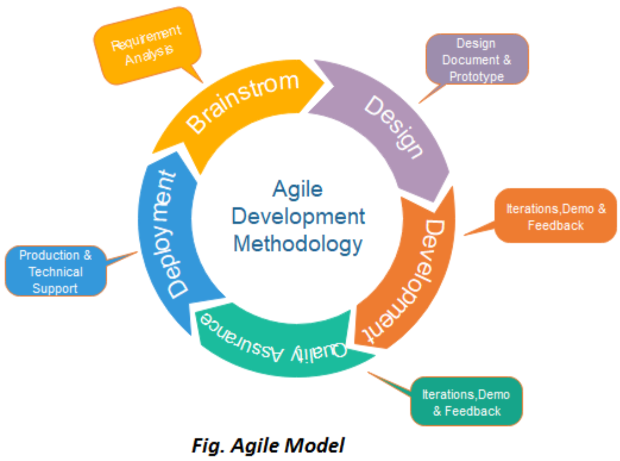


Figure 4.5: Agile Model

4.2.1 Data Collection & Prepossessing

Data Collection Firstly, we go to kaggle.com to collect the dataset. we search for Jewelry sales, Jewelry product lists or customer review datasets. Then download the Jewelry.csv dataset.

Prepossessing After collecting the dataset from Kaggle then import CSV file into Jupiter notebook. Then prepossess the data by -

- **Handling missing values**: Identify missing values in the dataset [as NaN, NULL, or other placeholders].
- **Data Splitting**: Split the dataset into training and test sets for model training.
- **Normalization**: Normalize data to ensure that different features have similar scales and distributions. Normalization techniques include min-max scaling (scaling features to a range between 0 and 1)

4.2.2 Algorithm

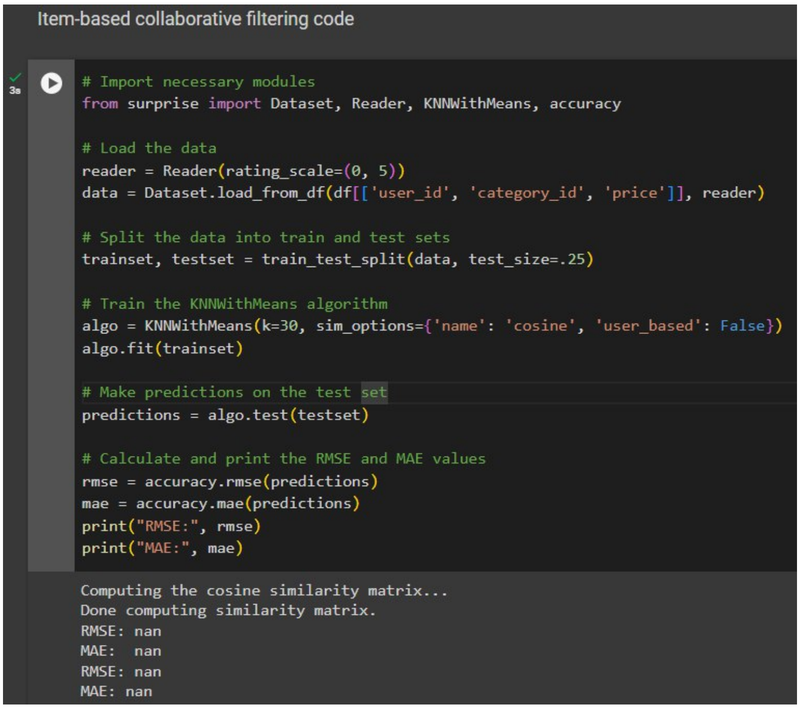
1. Instance-based Learning: The k-NN approach employs instance-based learning, where the model retains a memory of the complete training dataset.
2. Regression or Classification: It is applicable to jobs involving both regression and classification.
3. k Neighbors: The algorithm predicts the label or value of a new data point by considering the 'k' closest points in the training data.
4. Distance Metric: The distance between data points is commonly measured using the Euclidean distance, however alternative distance metrics may be employed.
5. the majority classification: The new data point is classified by allocating to it the most frequent class among its 'k' neighbors.
6. Regression: In regression, the average (or weighted average) of the 'k' neighbors' target values is determined.

7. Non-parametric: It does not make any assumptions about the underlying data distribution in non-parametric.

8. Lazy Learning: It's sometimes referred to as a lazy learner because it doesn't learn a model during training. It only memorizes the training data and performs the computation time of prediction.

9. Feature Scaling: Feature scaling is important as it's sensitive to the scale of features due to its reliance on distance metrics.

4.2.3 Recommendation Algorithms




```
# Import necessary modules
from surprise import Dataset, Reader, KNNWithMeans, accuracy

# Load the data
reader = Reader(rating_scale=(0, 5))
data = Dataset.load_from_df(df[['user_id', 'category_id', 'price']], reader)

# Split the data into train and test sets
trainset, testset = train_test_split(data, test_size=.25)

# Train the KNNWithMeans algorithm
algo = KNNWithMeans(k=30, sim_options={'name': 'cosine', 'user_based': False})
algo.fit(trainset)

# Make predictions on the test set
predictions = algo.test(testset)

# Calculate and print the RMSE and MAE values
rmse = accuracy.rmse(predictions)
mae = accuracy.mae(predictions)
print("RMSE:", rmse)
print("MAE:", mae)
```

```
Computing the cosine similarity matrix...
Done computing similarity matrix.
RMSE: nan
MAE:  nan
RMSE: nan
MAE: nan
```

Figure 4.6: item-based collaborative filtering

Related product recommendation code

```
# prompt: Related product recommendation code

def get_similar_products(product_id, df):
  product_category = df[df['category_id'] == df.loc[product_id, 'category_id']]['category'].iloc[0]
  similar_products = df[df['category'] == product_category]
  return similar_products

def recommend_related_products(product_id, df):
  similar_products = get_similar_products(product_id, df)
  recommendations = similar_products.loc[:, ['category_id', 'category', 'brand', 'price', 'type']]
  return recommendations.head()

product_id = 20000
recommendations = recommend_related_products(product_id, df)
print(recommendations)
```

```
     category_id         category  brand  price      type
1   1.806829e+18  jewelry.pendant    1.0  54.66  sapphire
2   1.806829e+18  jewelry.pendant    0.0  88.90   diamond
7   1.806829e+18  jewelry.pendant    2.0  60.27     pearl
19  1.806829e+18  jewelry.pendant    1.0  61.51    garnet
23  1.806829e+18  jewelry.pendant    4.0   1.03    fianit
```

Figure 4.7: Related product recommendation code

4.2.4 Summary

An e-commerce model that combines sophisticated machine learning algorithms, a user-friendly interface, and an interactive chatbot is described in Chapter 3. The concept seeks to improve user experience and security. Important features include a chatbot for individualized support, an intuitive user interface for simple navigation, and a fraud detection system that leverages historical transaction data. The chapter also adds a feasibility analysis covering technical, operational, economic and schedule constraints, affirming the model's potential to revolutionize the e-commerce landscape.

Chapter 5

5 Implementation and Testing

5.1 Introduction

We will discuss in this section the step-by-step implementation of our e-commerce project. It also discusses the hardware and software requirements of setting up a functional e-commerce website, as well as the step-by-step process of installing, configuring, coding and integrating all the various system components.

5.2 Hardware Requirement

Operating System: Windows 10, 11 (64-bit), Linux

Processor : AMD Ryzen 7 4700U with Radeon Graphics, 2.00 GHz.

Code Editor : VSCode or PyCharm

RAM : 4/8 GB

SSD : 126GB

Storage : 10GB-30GB

5.3 System Requirement

System Type : Web Application

Database : SQLite3

Front-end : Python, Html, CSS, JS, Ajax

Back-end : Django Framework

Python Version : 3.12.2

Django version : 5.0.3

Pip version : 24.0

5.3.1 E-commerce System

An e-commerce web application provides solutions and services to customers using integrated tools like search products, product display tools, shopping carts, payment systems etc.

5.4 Database Server and Administration tools

we offer comprehensive solutions tailored to meet the specific needs of any online store.

1. Optimized Database Server: Our e-commerce site can handle a high volume of transactions seamlessly.

2. Data Security: We understand the importance of protecting our customer's sensitive information. Our administration tools include advanced security measures to safeguard your database against unauthorized access.

3. Backup and Recovery: Never worry about losing valuable data again. Our system includes automated backup and recovery features.

4. Monitoring and Performance: Stay on top of your database's performance with real-time monitoring tools. Identify and address potential bottlenecks to optimize the speed and efficiency of your online store.

5. User-friendly Interface: Our administration tools are designed with simplicity and easy to use. Any database management fresher can handle our system database.

5.5 System UI

Here we can see the user interface, user sign up, log in, password reset or forgot password, user order items, view items, related product section, user add to cart, user customize profile, change password, Then admin login s ection. Here admin can create a user or admin and also manage website data. Providing our E-commerce shops screenshots below:

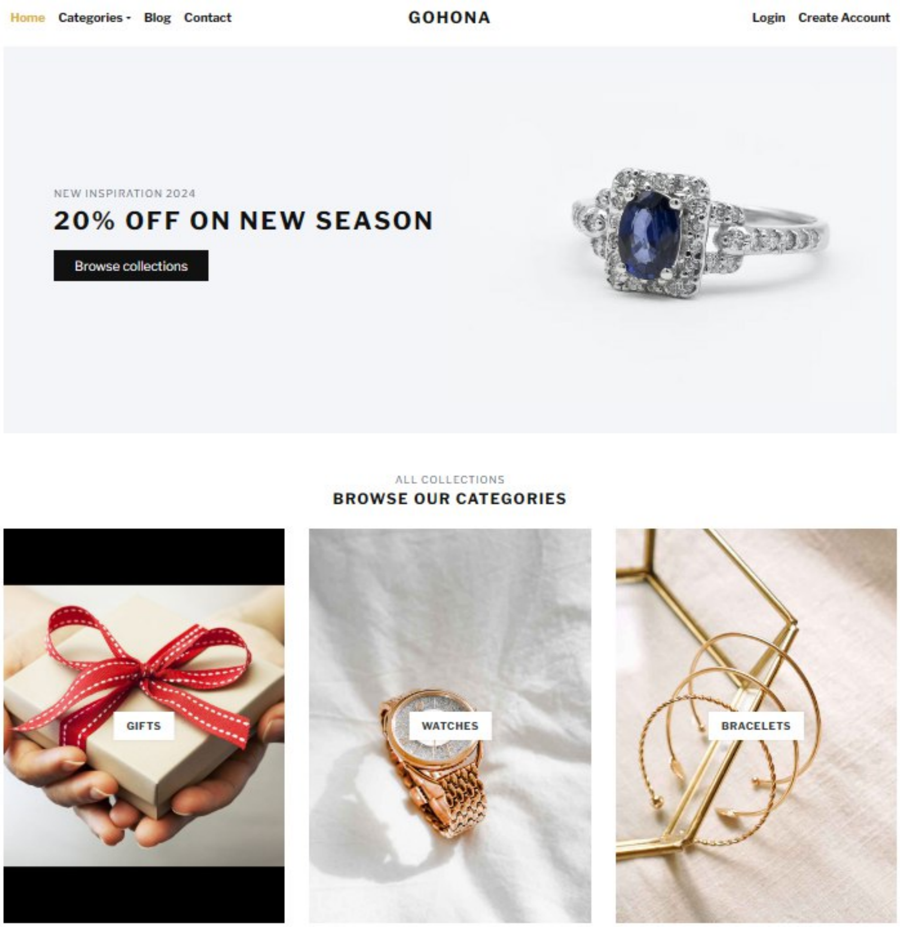


Figure 5.8: user-interface 1

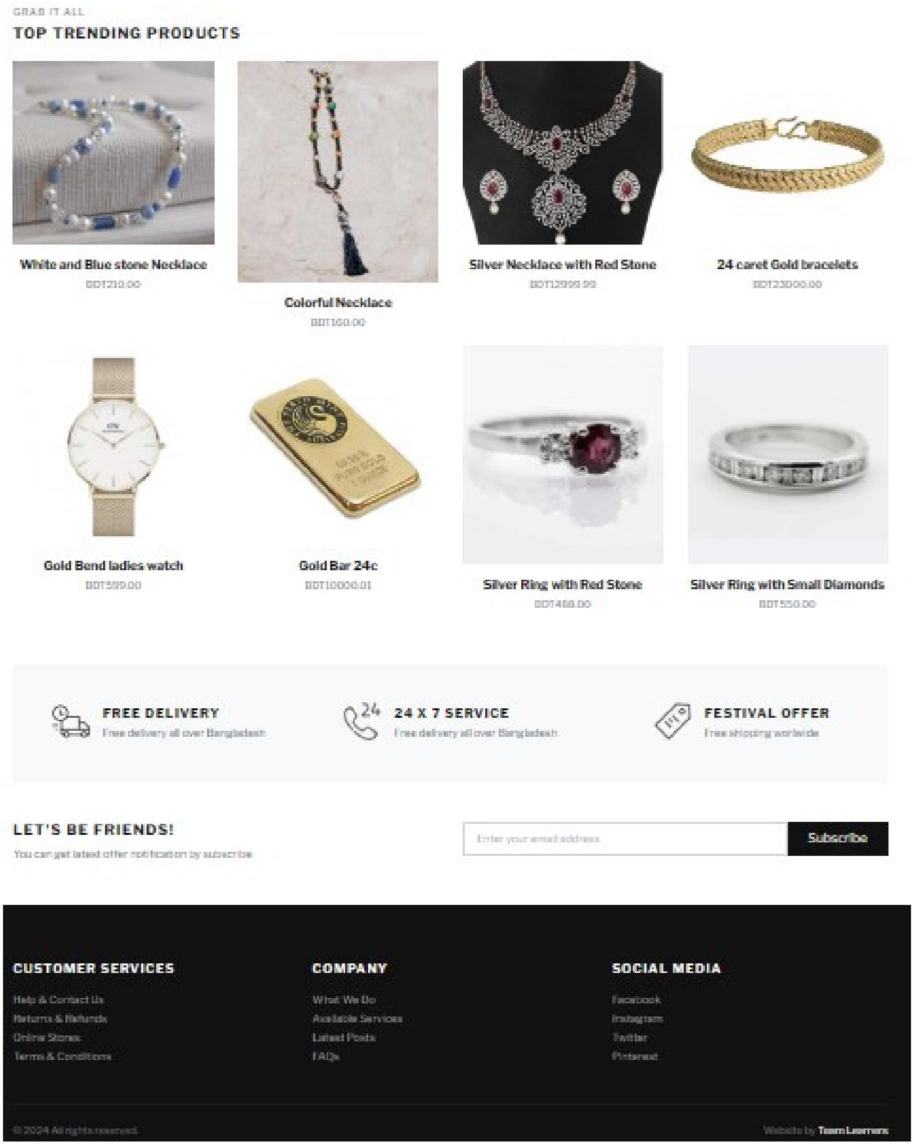


Figure 5.9: user-interface 2

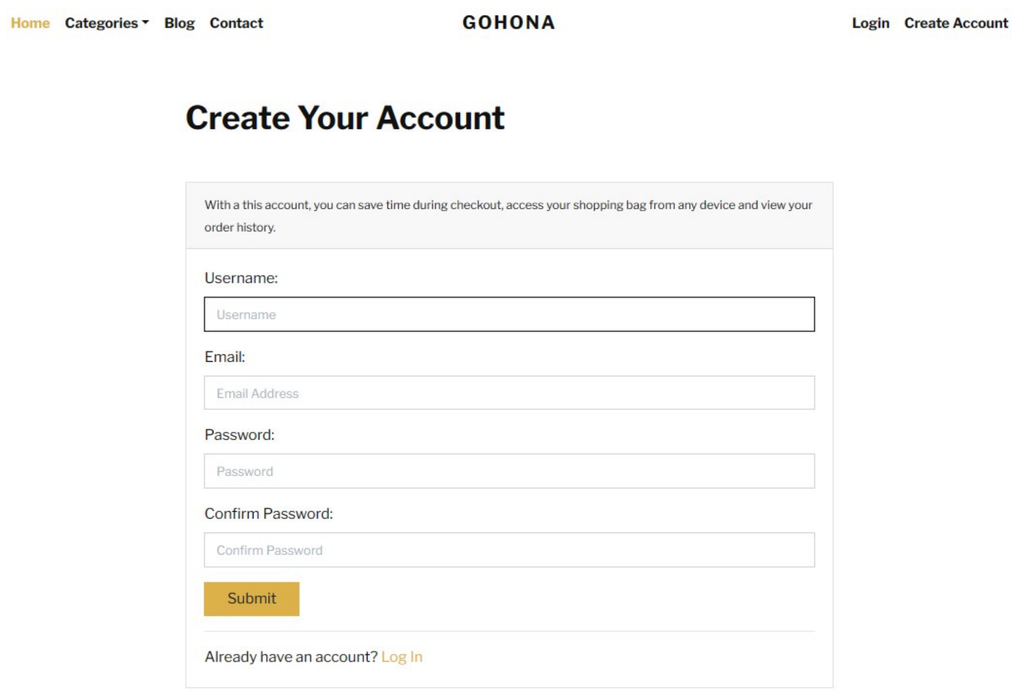


Figure 5.10: user-create account

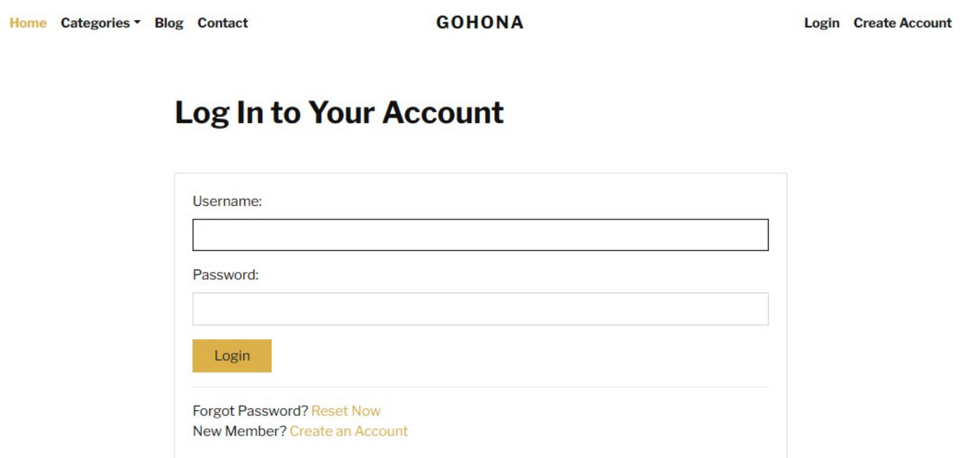


Figure 5.11: user-login account

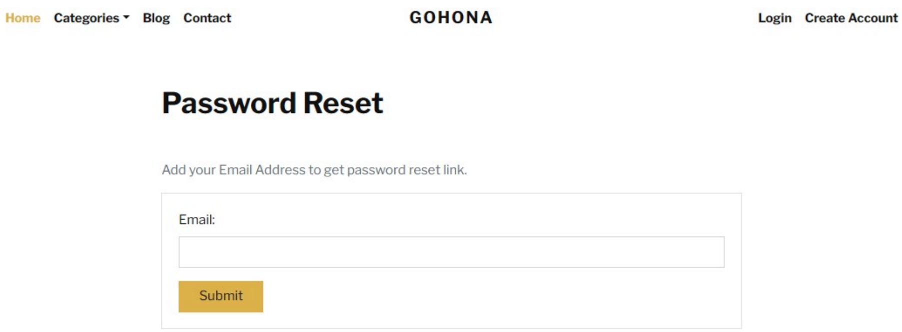


Figure 5.12: user-reset password

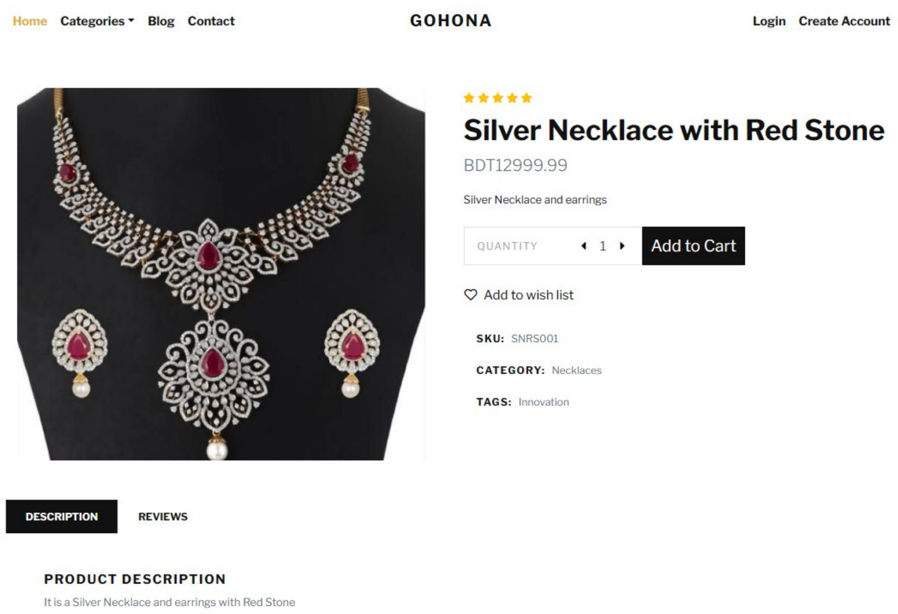


Figure 5.13: user -item

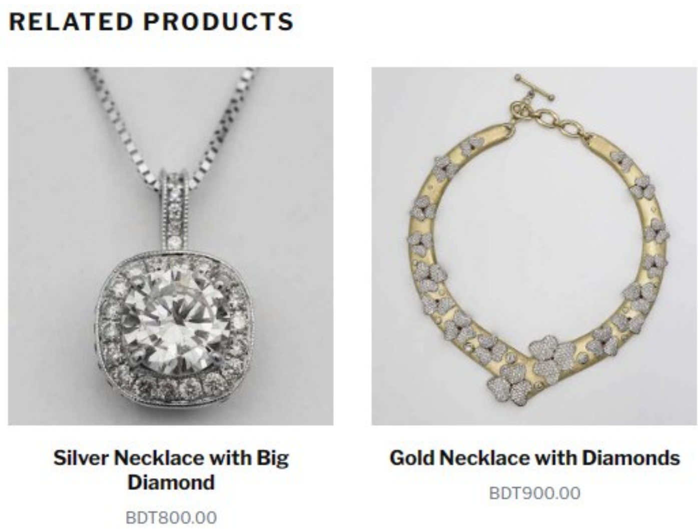


Figure 5.14: user-related product

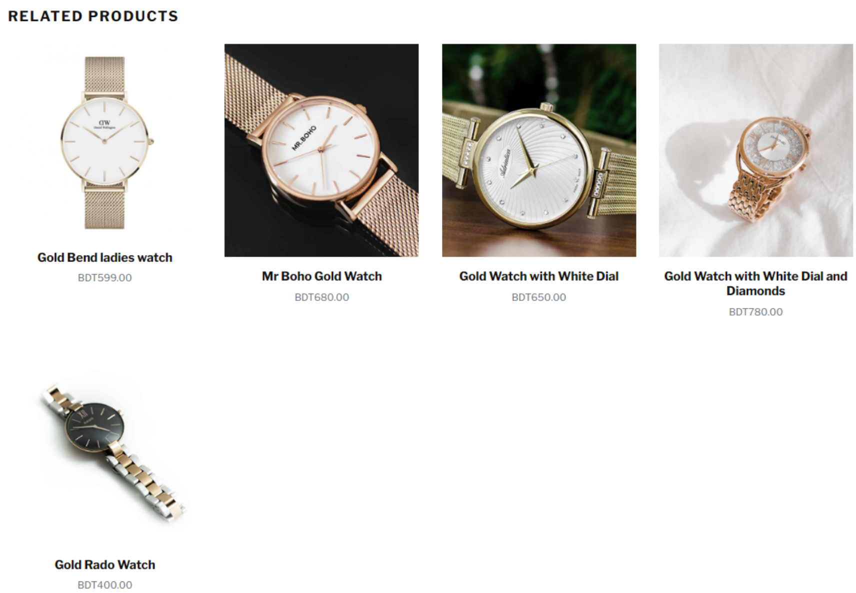


Figure 5.15: Related product of watch category

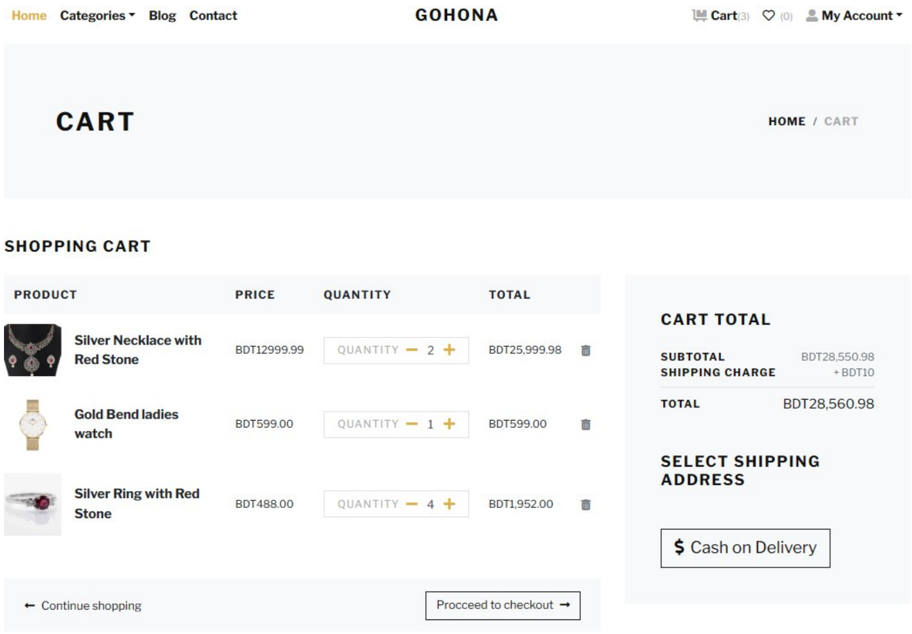


Figure 5.16: user-add to cart

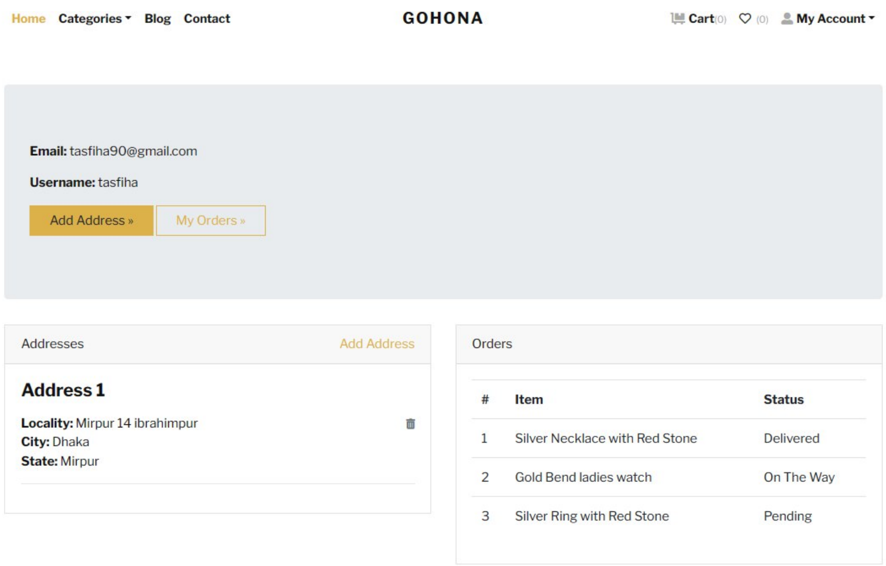


Figure 5.17: user-add address

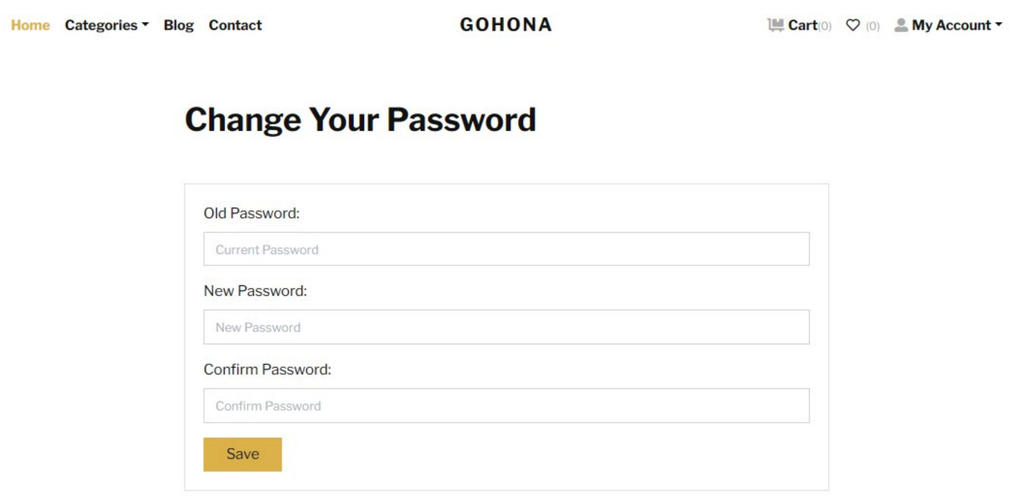


Figure 5.18: user-change password

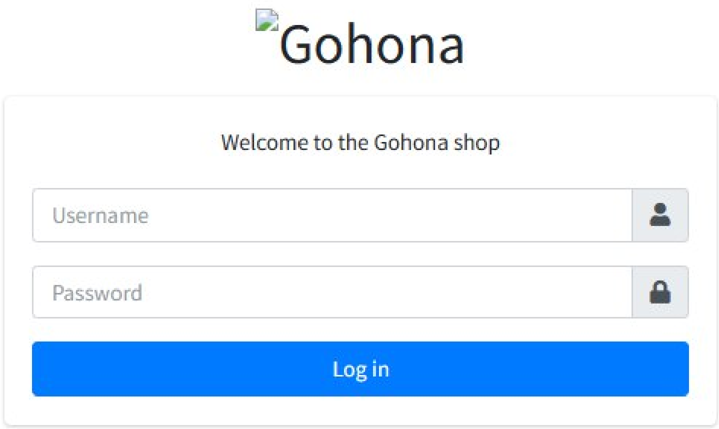


Figure 5.19: Admin login

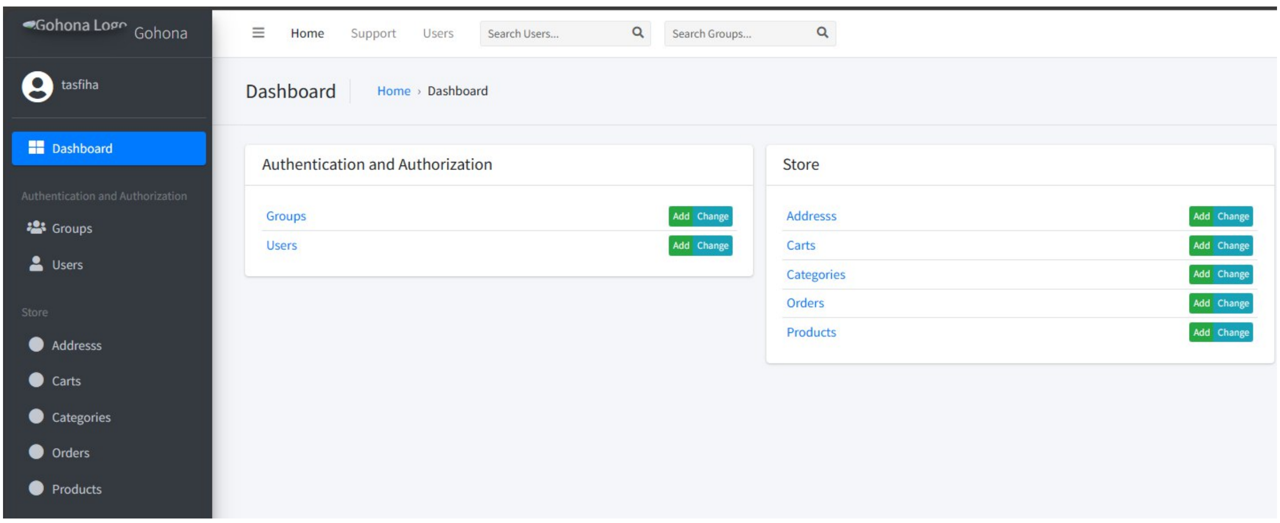


Figure 5.20: admin panel-dashboard

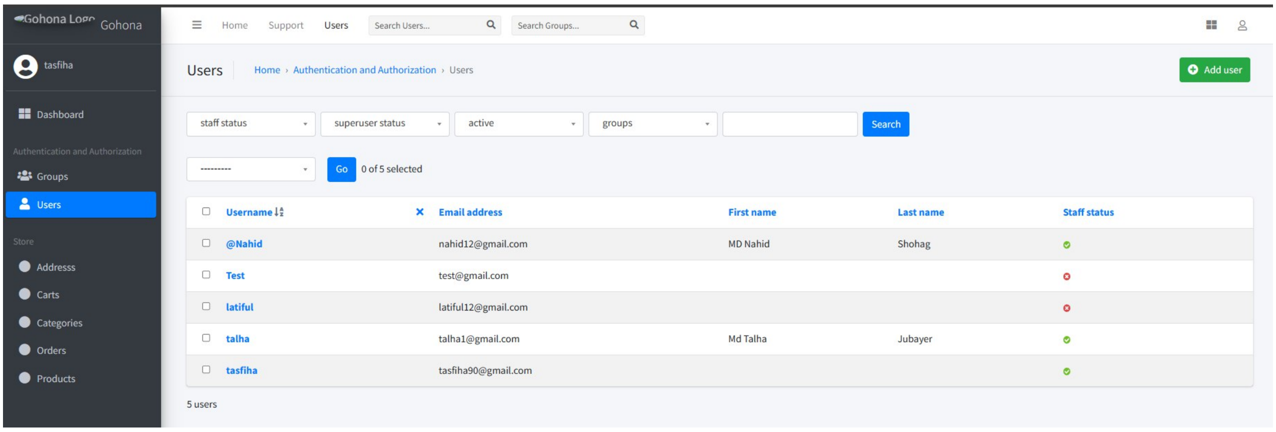


Figure 5.21: admin panel-user list

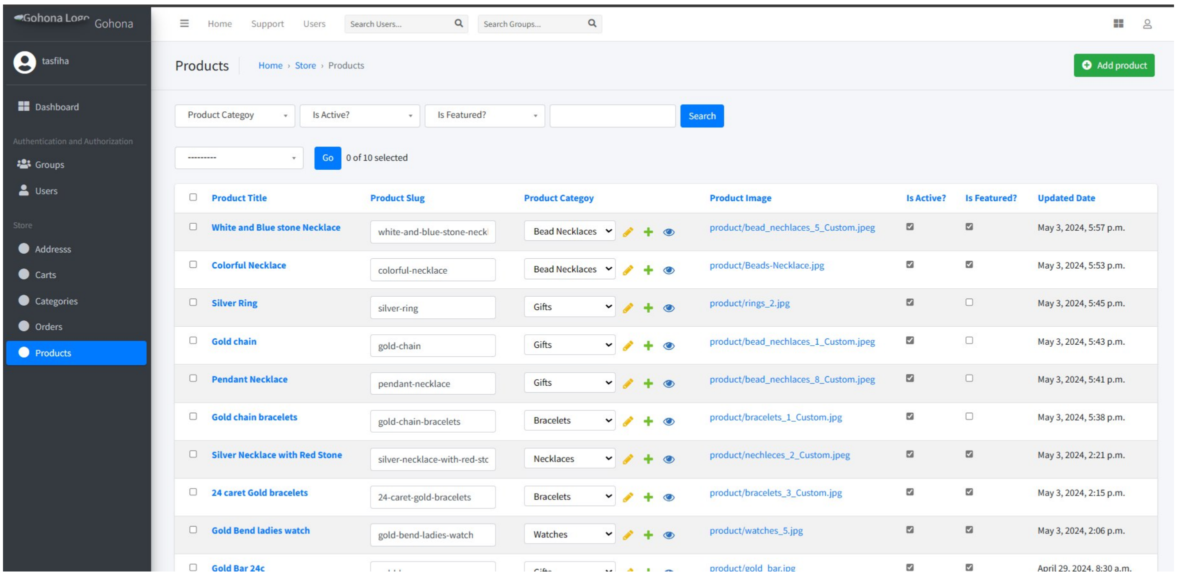


Figure 5.22: admin panel-product list

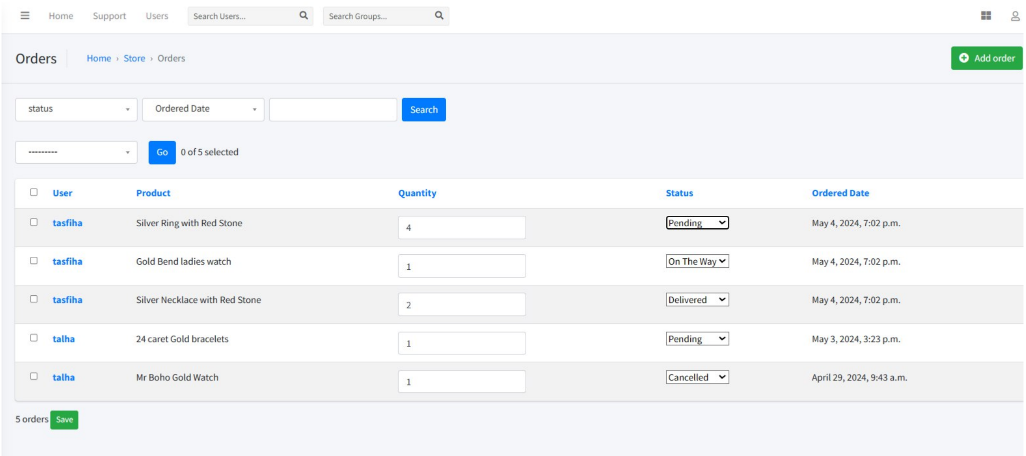


Figure 5.23: admin panel-order list

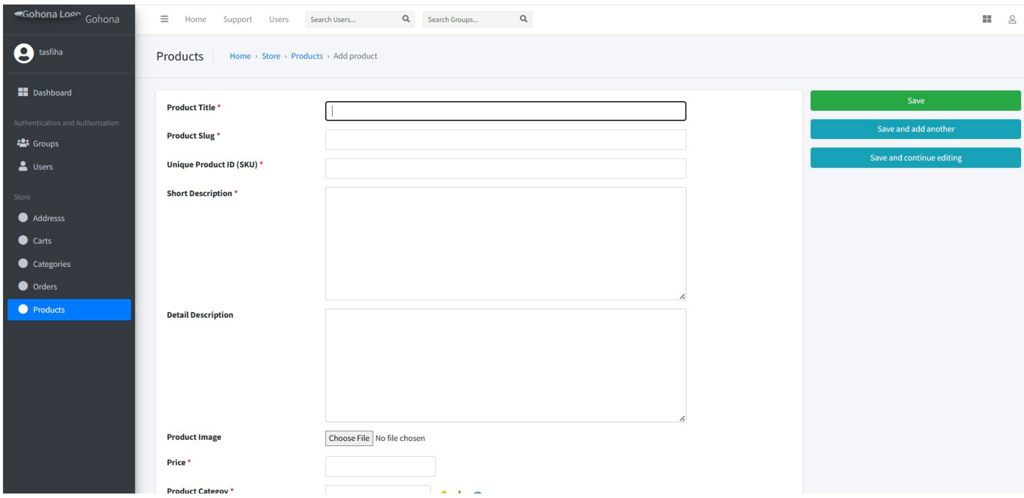


Figure 5.24: admin panel-add product

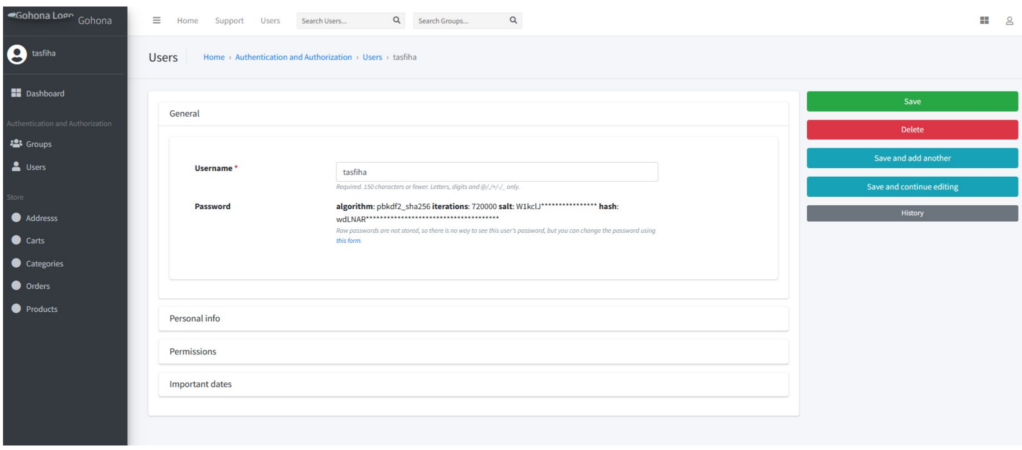


Figure 5.25: admin panel-user verification

Chapter 6

6 Conclusion and Future Enhancement

6.1 Conclusion

Our innovative e-commerce platform blends advanced technology and user-centered design to reshape online shopping. By seamlessly incorporating advanced machine learning, we enhance security and user engagement, with product recommendations and a user-friendly interface. An intelligent chatbot will be added to our website for a customer support experience. Committed to security, user experience, and innovation, our platform is ready to transform online shopping into a new era of dynamic, secure experiences.

6.2 Limitations

We have some limitations in our project which are-

- Payment Gateway,
- Chatbot's Integration and
- Custom Inventory Admin Dashboard.

We faced some system issues which is why we could not develop the Payment Gateway, Chatbot's Integration, and Custom Inventory Admin Dashboard

6.3 Future Enhancement

In the future, we will solve our system issue and develop an AI chatbot for 24/7 Customer Support, a custom admin UI for a more user-friendly interface and integrate SSLCOMMERZ Payment Gateway into our project.

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